

$\mathbb{Q}$ ,  $\mathbb{R}$  &  $\mathbb{C}$  are fields. What differentiates them?

$$\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

Order: we say that a field is an ordered field when  
 $\hookrightarrow +, \cdot$

1) For any  $x, y$  in the field, either  $x < y$  or  $x > y$  or  $x = y$ .

2)  $x < y$  &  $y < z \Rightarrow x < z$  Transitivity

3) If  $x < y$  then  $x + z < y + z$  for any  $z$ .

Order axioms

4) If  $x < y$  &  $z > 0$  then  $xz < yz$

Exercise: ~~Prove:~~ If  $0 < c < d$  then  $ac < bd$ ,  
 $0 < a < b$

$\mathbb{Q}$  &  $\mathbb{R}$  are ordered fields, while  $\mathbb{C}$  is not.

The distinguishing aspect of  $\mathbb{Q}$  &  $\mathbb{R}$  is "completeness".

$\hookrightarrow$  No HOLES

Completeness can be defined in multiple ways

① Sequences:

$$(a_n)_{n \in \mathbb{N}} \rightarrow \mathbb{Q}, \mathbb{R}.$$

We say that a sequence  $a_n$  has a limit  $a$  if for any

$\epsilon > 0$ ,  $\exists N$  such that

$$|a_n - a| < \epsilon \text{ for all } n \geq N.$$

$\hookrightarrow$   
how far away  
from limit

$\rightarrow a_n$  within  $\epsilon$  of  $a$

Example:

Consider

$$a_n = \frac{1}{n}$$

$$\lim a_n = 0.$$

Let  $\epsilon > 0$  be given. Pick

$$N = \left\lceil \frac{1}{\epsilon} \right\rceil$$

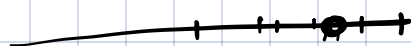
$$\left| \frac{1}{n} - 0 \right| < \epsilon$$

for all  $n \geq N$

then for all  $n \geq N$ , we can see that

$$|a_n - a| = \left| \frac{1}{n} - 0 \right| < \frac{1}{\left\lceil \frac{1}{\epsilon} \right\rceil} < \epsilon.$$

$$\text{So } \lim a_n = 0.$$



Example:

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$$

$$a_1 = 1$$

$$a_4 = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16}$$

$$\rightarrow \frac{\pi^2}{6}$$

$$a_2 = 1 + \frac{1}{4}$$

$$a_3 = 1 + \frac{1}{4} + \frac{1}{9}$$

We say a field is complete if for every sequence  $a_n$  in the field with  $\lim a_n = a$ ,  $a$  is also in the field.

$\mathbb{Q}$  is incomplete,  $\mathbb{R}$  is complete.

② Supremum:

We say that  $u$  is an upper bound of a set  $S$  if  $u \geq s$  for every  $s \in S$ .

We say that  $p$  is the least upper bound of  $S$

if  $p \leq u$  for any  $u$  that is an upper bound of  $S$ .

$$\text{Ex 1} \quad (0, 1)$$

$$\text{If } S = (0, 1)$$

$$\sup S = 1$$

$\max S$  not defined.

$$\text{Ex 2} \quad \underbrace{(0, 1]}_{S = (0, 1]}$$

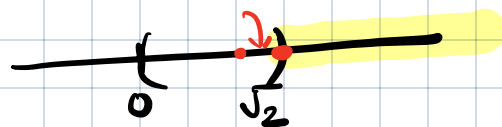
$$S = (0, 1]$$

$$\sup S = 1 = \max S$$

If the set has a max, it is equal to sup.

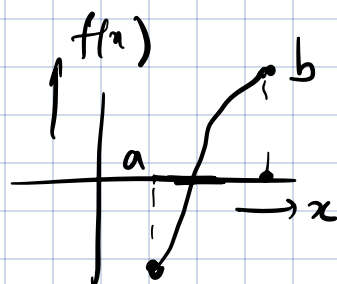
$$\text{Ex 3: } S = (0, \sqrt{2}) \cap \mathbb{Q}$$

Another way to state completeness:  
Every set bounded above must have a supremum in the field



a supremum in

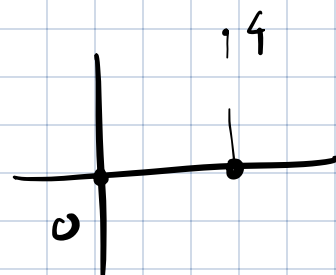
③ Intermediate value theorem



$$f: \mathbb{Q} \cap [0, 2] \rightarrow \mathbb{Q}$$

defined as

$$f(x) = x^2$$

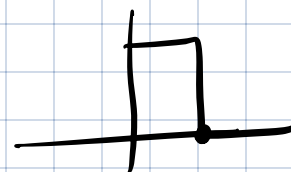


Continuous  $f^n$ :

for every  $\epsilon > 0$ , there exists  $\delta > 0$  such that

$$|x - c| < \delta \Rightarrow |f(x) - f(c)| < \epsilon.$$

then we say  $f$  is continuous at  $c$ .



of real numbers

Completeness: Every set that is bounded above has a supremum,

10 15 -1 -2 3

$\mathbb{R}$  is a complete ordered field

$(0, 1)$

$(0, \infty)$

Property 1: Sandwich theorem

$$a_n \leq b_n \leq c_n \quad \text{for all } n.$$

$$\lim a_n = \lim c_n = l$$

$$\text{then } \lim b_n = l.$$

Property 2: Suppose  $(a_n)$  is a sequence of real numbers that is monotonic and bounded above:

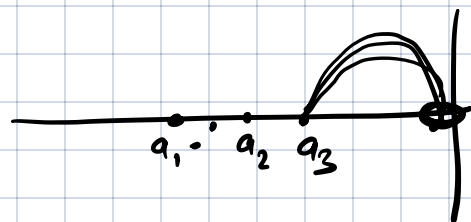
$$a_n \leq a_{n+1} \quad \text{for all } n$$

$$a_1 \leq a_2 \leq a_3 \leq a_4 \dots$$

$$a_n \leq M \quad \text{for all } n \quad \text{for some } M.$$

$$1 - \frac{1}{n} \quad 1 - e^{-n}$$

then  $a_n$  converges.



Proof: Consider  $S = U(a_n)$

then  $S$  is bounded above, and so must have a supremum.

$$l = \sup S.$$

$$l - \epsilon < a_n$$

$\epsilon$

$$\lim a_n = l$$



$$\epsilon > 0$$

Suppose  $\epsilon > 0$   
is given.

$$|a_n - l| < \epsilon$$

for all  $n > N$

Since  $l$  is a supremum,  $\exists$  an  $N$

such that  $a_n > l - \epsilon$

then by monotonicity,

$$a_n > l - \epsilon \text{ for all } n \geq N$$

thus  $l - \epsilon < a_n < l + \epsilon$  for all  $n \geq N$ .

showing that  $\lim a_n = l$ .

Recap:

① We say that a sequence  $a_n$  converges to  $a$  ( $a_n \rightarrow a$ )



for any  $\epsilon > 0$ , there exists  $N$  such that

$$|a_n - a| < \epsilon \text{ for all } n \geq N.$$

② We say that a sequence diverges to infinity ( $a_n \rightarrow \infty$ )

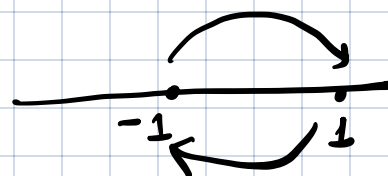
for any  $m > 0$ , there exists  $N$  such that

$$a_n > m \text{ for all } n \geq N.$$

All sequences do not necessarily have a limit.

$$a_n = (-1)^n$$

$$a_n = (-1)^n n$$



Tool 1: Squeeze theorem

$$\left. \begin{array}{l} a_n \leq b_n \leq c_n \text{ be sequences.} \\ \lim a_n = \lim c_n = a \\ \text{then } b_n \rightarrow a. \end{array} \right\}$$

Optional:

Can be proved using definition of limit.

Tool 2: Binomial theorem

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots$$

If  $n$  is integer

If  $x > 0$  then

$$(1+x)^n \geq 1+nx \quad \leftarrow$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Example: For any  $a > 0$ , show that

$$\lim_{n \rightarrow \infty} a^{1/n} = 1$$

$$\text{Let } b_n = a^{1/n} - 1$$

Goal:  $b_n \rightarrow 0$

$$e^{\ln a^{1/n}} = e^{\frac{1}{n} \ln a}$$

$$(1+b_n)^n = a$$

$$1+nb_n \leq \underbrace{1+nb_n + \binom{n}{2}b_n^2 + \dots}_{=a} = a$$

$$\text{Case 1 } \left( \begin{array}{l} a > 1 \\ \Rightarrow b_n > 0 \end{array} \right)$$

$$1+nb_n \leq a$$

$$\Rightarrow 0 \leq b_n \leq \underbrace{\frac{a-1}{n}}$$

Since  $\frac{a-1}{n} \rightarrow 0$ , we have by Squeeze theorem,  $b_n \rightarrow 0$ .

If  $a < 1$   $a^{\frac{1}{n}} = \frac{1}{a^{1/n}}$  where  $a = \frac{1}{a}$ ,  $a > 1$

Since  $\lim_{n \rightarrow \infty} a^{\frac{1}{n}} = 1$ , it follows that  $\lim_{n \rightarrow \infty} \frac{1}{a^{1/n}} = 1$ .

Example 2:

$$\lim_{n \rightarrow \infty} \frac{1}{n^a} = \begin{cases} 0 & a > 0 \\ \infty & a < 0 \\ 1 & a = 0 \end{cases}$$

requires a proof.

Example 3:

$$\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$$

(Exercise)

Outline:  $b_n = n^{\frac{1}{n}} - 1$

Goal:  $b_n \rightarrow 0$

$$(1 + b_n)^n = n$$

$$n C_2 b_n^2 \leq 1 + n b_n + n C_2 b_n^2 + \dots = n$$

$$\frac{n^2}{1000} > n$$

$$b_n^2 \leq \frac{n}{n C_2} = \frac{2n}{n(n-1)} = \frac{2}{n-1}$$

$$0 \leq b_n^2 \leq \frac{2}{n-1}$$



Tool 3: Continuity:

If  $f$  is continuous at  $a$ , then  
for every sequence  $a_n \rightarrow a$   
we have  $f(a_n) \rightarrow f(a)$ .

Recap:

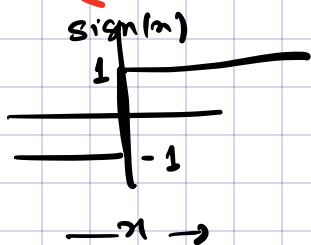
We say  $f$  is continuous at  $a$

if for every  $\epsilon > 0$

there exists  $\delta > 0$

such that

(LOCAL)



$$|x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon.$$

$$a_n \rightarrow a$$

does  $\text{sign}(a_n) = \begin{cases} 1 & \text{if } a_n > 0 \\ -1 & \text{if } a_n < 0 \end{cases}$

converge?

does  $a_n^2 \rightarrow a^2$

$$f(x) = x^2$$

$$e^{a_n} \rightarrow e^a$$

$a > b$ , then

$$\text{sign}(a_n) \rightarrow \text{sign}(a).$$

of

$$a_n = \underbrace{\frac{(-1)^n}{n}}$$

$$\lim a_n = 0$$

$$\text{sign}(a_n) = (-1)^n$$

does not converge.

$$\begin{cases} \lim e^{\frac{1}{n}} = 1 \\ \lim a^{\frac{1}{n}} = 1 \end{cases}$$

$$f(x) = e^x$$

$$f(x) = a^x$$

$$\lim \sin \frac{1}{n} = 0$$

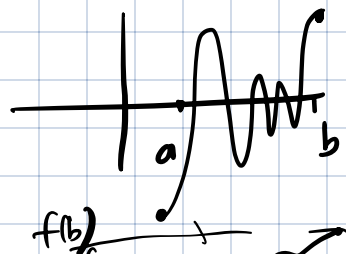
Tool 4: Intermediate value theorem.

$f: [a, b] \rightarrow \mathbb{R}$  be continuous

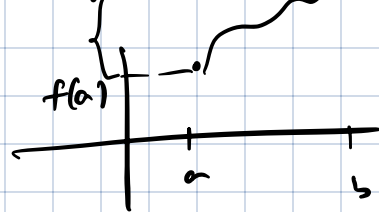
if  $f(a) < 0$  &  $f(b) > 0$  then there exist  $c \in [a, b]$

such that  $f(c) = 0$ .

closely connected to completeness.







Tool 5: Sums & products of functions.

$$\lim_{n \rightarrow \infty} a_n b_n$$

$$\lim_{n \rightarrow \infty} (a_n + b_n)$$

$$a_n = n \quad b_n = d - n$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

provided the limit  $a_n + b_n$  exists.

then  $a_n \rightarrow \infty, \quad b_n \rightarrow -\infty$

$$a_n + b_n = d$$

$$\infty - \infty, \quad 0 \cdot \infty, \quad \frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 1^\infty, \quad \infty^0$$

Tool 6: L'Hopital's rule:

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$$

$$\lim_{x \rightarrow c} f(x) = 0$$

$$= \lim_{x \rightarrow c} g(x)$$

$\lim_{x \rightarrow c} f(x) = l$   
for any  $\epsilon > 0$ , there exists  $\delta > 0$  such that  
 $|x - c| < \delta \Rightarrow |f(x) - l| < \epsilon$

if  $f$  &  $g$  are differentiable around  $c$ ,

and (i)  $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$  exists

(ii)  $g'(x) \neq 0$  around  $c$

then  $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$

Examples:

$$\lim_{n \rightarrow \infty} n e^{-n} = \lim_{n \rightarrow \infty} \frac{n}{e^n} = \lim_{n \rightarrow \infty} \frac{1}{e^n} = \lim_{n \rightarrow \infty} \frac{1}{e^n} = 0$$

$$\lim_{x \rightarrow \infty} x^2 e^{-x} = \lim_{x \rightarrow \infty} \frac{x^2}{e^x} = \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0$$

$$\frac{2x}{e^x} \quad \frac{2}{e^x}$$

$$\lim_{n \rightarrow \infty} x^n e^{-x} = 0$$

→ No polynomial can grow faster than exponential.

$$(x^n + 2x^{n-1} + \dots + 1)e^{-x}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$$

$$f(x) = \left(1 + \frac{1}{x}\right)^x$$

$$\log f(x) = x \log\left(1 + \frac{1}{x}\right)$$

Apply L'Hôpital's rule to find  $\lim_{x \rightarrow \infty}$ .

$1^\infty$  indeterminate

resolves the  $1^\infty$  discrepancy

$$\left(1 + \frac{1}{n}\right)^n = e$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{\sqrt{n}}\right)^n = \left(\left(1 + \frac{1}{\sqrt{n}}\right)^{\sqrt{n}}\right)^{\sqrt{n}} \rightarrow \infty$$

$$\left(1 + \frac{1}{n}\right)^{n^2} \rightarrow \infty$$

$$\left(\underbrace{\left(1 + \frac{1}{n}\right)^n}_e\right)^n$$

$$\lim \left(1 + \frac{1}{n^2}\right)^n = \lim \left( \underbrace{\left(1 + \frac{1}{n^2}\right)^{n^2}}^{\frac{1}{n}} \right)^{\frac{1}{n}} = \lim e^{\frac{1}{n}} = 1.$$

$$\lim \underbrace{\left(1 + \frac{1}{n^{0.99}}\right)^n}_{\rightarrow \infty}$$

$$\lim \underbrace{\left(1 + \frac{1}{n^{1.01}}\right)^n}_{\rightarrow 0}$$

$$\lim \left(1 + \frac{2}{n}\right)^n = e^2$$

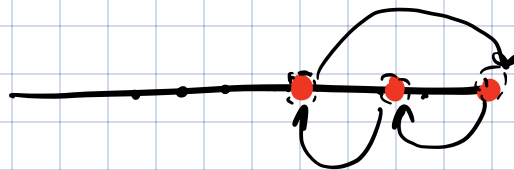
$$\lim \left(1 - \frac{1}{n}\right)^n = e^{-1}$$

Non-convergent Sequences.

$$\underbrace{(-1)^n}$$

$$\underbrace{(-1)^n n}$$

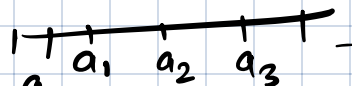
limit point: We say that  $l$  is a limit point of  $a_n$  if for any  $\epsilon > 0$ ,  $|a_n - l| < \epsilon$  for infinitely many  $a_n$ .



Suppose a sequence is bounded, i.e.

$$|a_n| < M \text{ for all } n.$$

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup \{a_k \mid k \geq n\}$$



$$\liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf \{a_k \mid k \geq n\}$$

$$\begin{array}{ccccccc} 1 & 5 & 3 & -2 & -4 & \sqrt{2} & \sqrt{5} & - & - \\ n=1 & n=2 & n=3 & & & & & & \end{array}$$

$$\overbrace{5 \quad 5 \quad 3 \quad \sqrt{5} \quad \sqrt{5} \quad \sqrt{5}}$$

$\sup \{a_k : k \geq n\}$  is a decreasing sequence  $\leftarrow$

$\inf \{a_k : k \geq n\}$  is an increasing sequence.

$$\sup \leftrightarrow \max$$

$$\inf \leftrightarrow \min.$$



$$\limsup_{n \rightarrow \infty} (-1)^n$$

$$-1, 1, -1, 1, -1, 1, \dots$$

Sequence  $a_n$  converges if & only if  $\limsup a_n = \liminf a_n$